

Module 8 - Design Strategy

Creating a blueprint for the system

Once we **know the design** for the new or improved system, we can choose whether to **build the system, buy an existing solution or have someone make the system for us.**

How does a **blueprint (for a system)**
look like?

The Design Phase

- Recommended System Acquisition Strategy
- System Acquisition Weighted Alternative Matrix
- Architecture Design
- Hardware and Software Specification
- Interface Design
- Physical Process Model
- Program Design Specifications
- Physical Data Model
- Data Storage Design
- Updated CRUD Matrix
- Updated CASE Repository Entries



- **Decide** whether to build, buy or outsource the system.
- What was that decision based on? Show us the **comparison of possible options**.
- Arrange system components to develop an **architecture**.
 - What **hardware** and **software** will be used?
 - How might the **interface** look like?
 - Physical **Process Model** ?
 - Code Structure ?
 - Database Structure / Physical Data Model ?
 - Database Architecture & Performance ?

FIGURE 6-2 System specification outline.

System Acquisition

Should we build, buy or outsource?

Primary Approaches

1. Develop a **custom application in-house**
2. **Buy** a packaged system and (possibly) **customize** it
3. Rely on an **external vendor, developer, or service provider** to build or provide the system

Custom Development

- Get exactly what we want
- The new system will be **consistent** with the technology and standards we are already using.
- Custom development enables us to **build and retain technical skills and functional knowledge** in-house
- Allows team **flexibility and creativity**
- The **unique solution** might provide a strategic advantage

Drone Sales Process

- **Business Requirement:** A web-based sales capability that enables customers to *select and customize the drone for purchase*.
 - A sales process alone is a standardized process.
 - However, DronTeq has a **unique functional requirement**: the **ordering process should link to a shop management process** that evaluates the customer's selected drone configuration to make sure it is an acceptable configuration.

Custom Order Approval Process

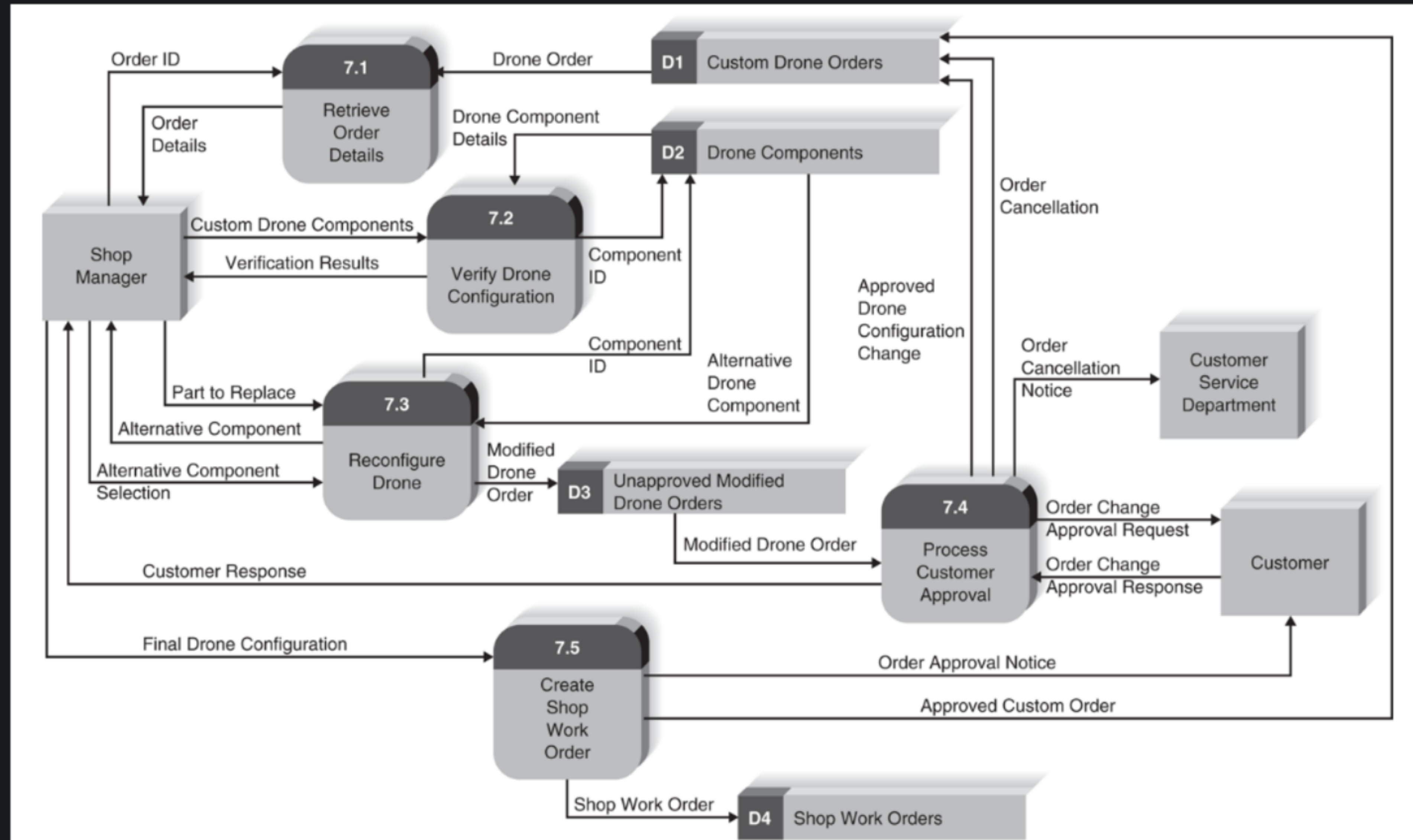
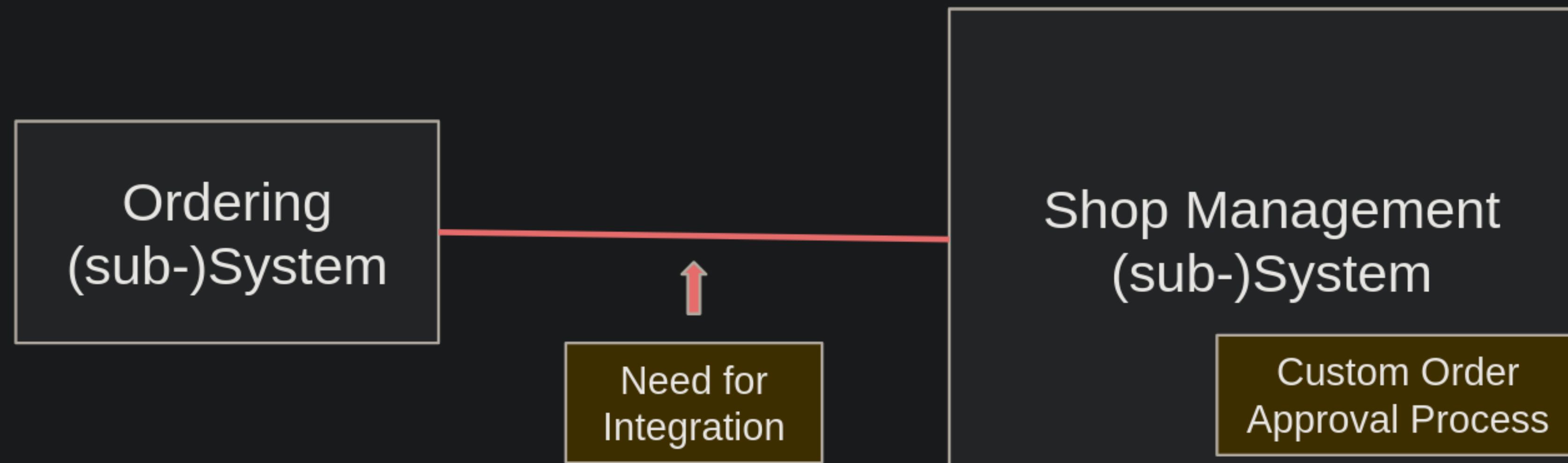


FIGURE 4-12 Shop manager approval of custom drone order level 1 DFD.

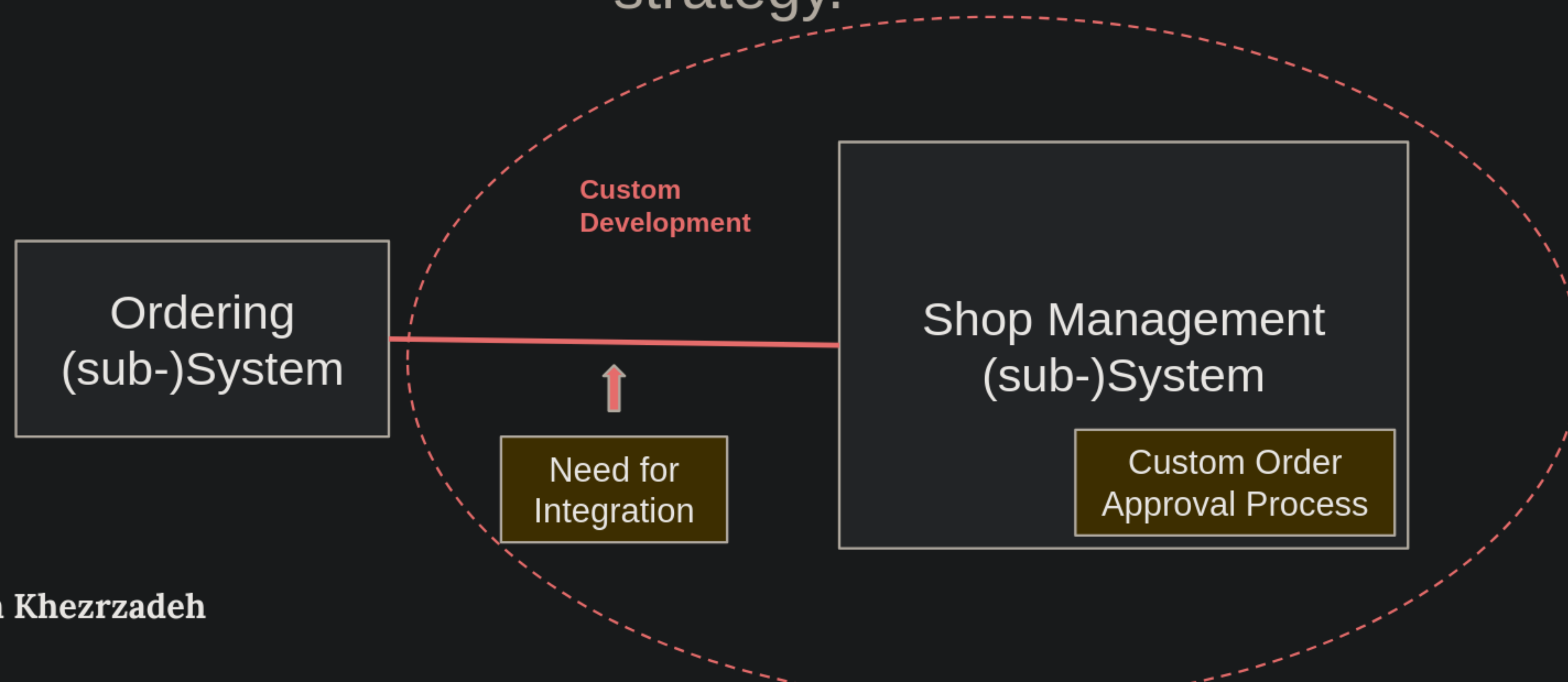
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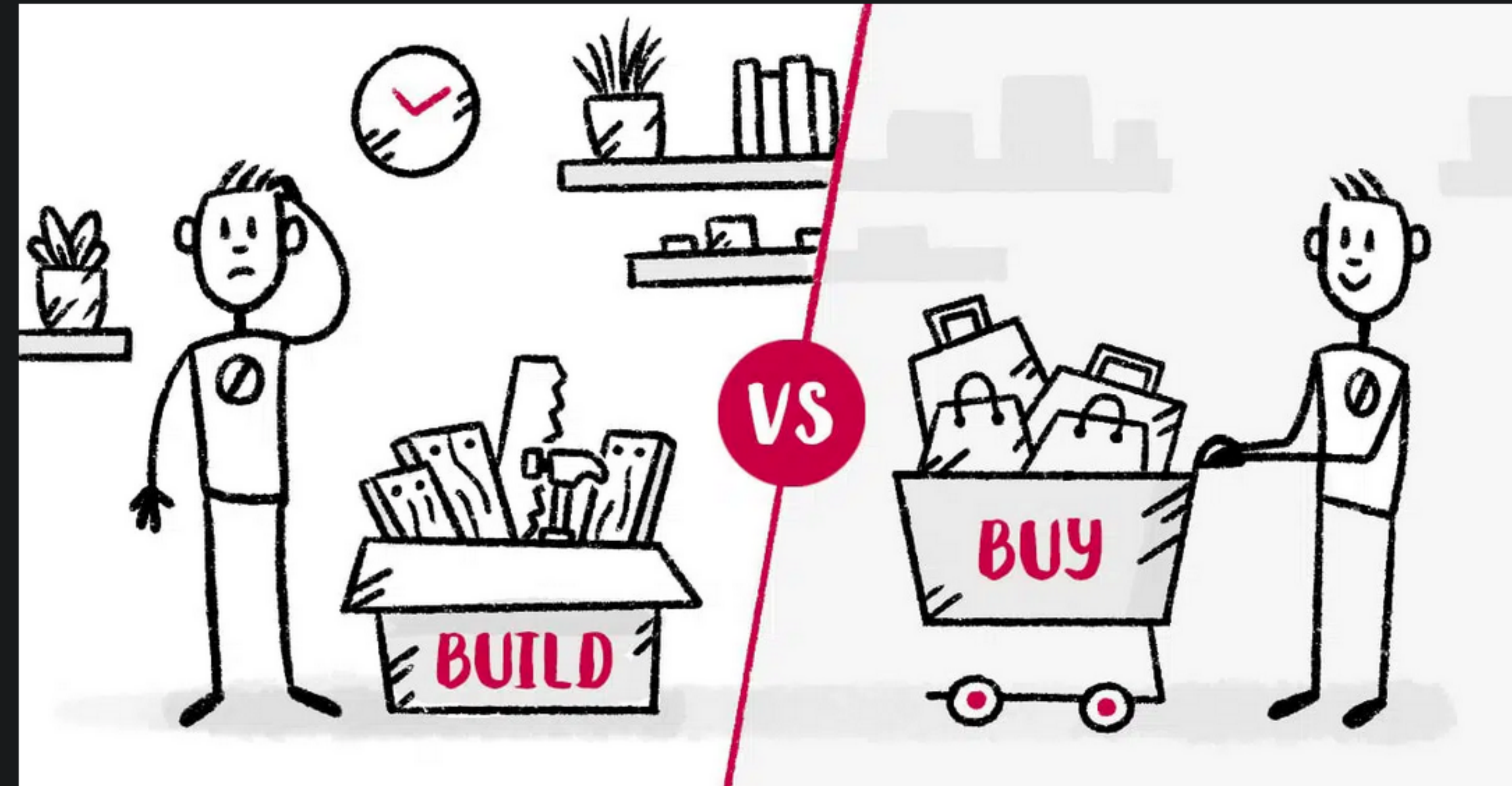
Drone Sales Process

The **need for integration** and the **unique approval process** are complex and highly specialized. Suggesting that **custom development** might be a good strategy.



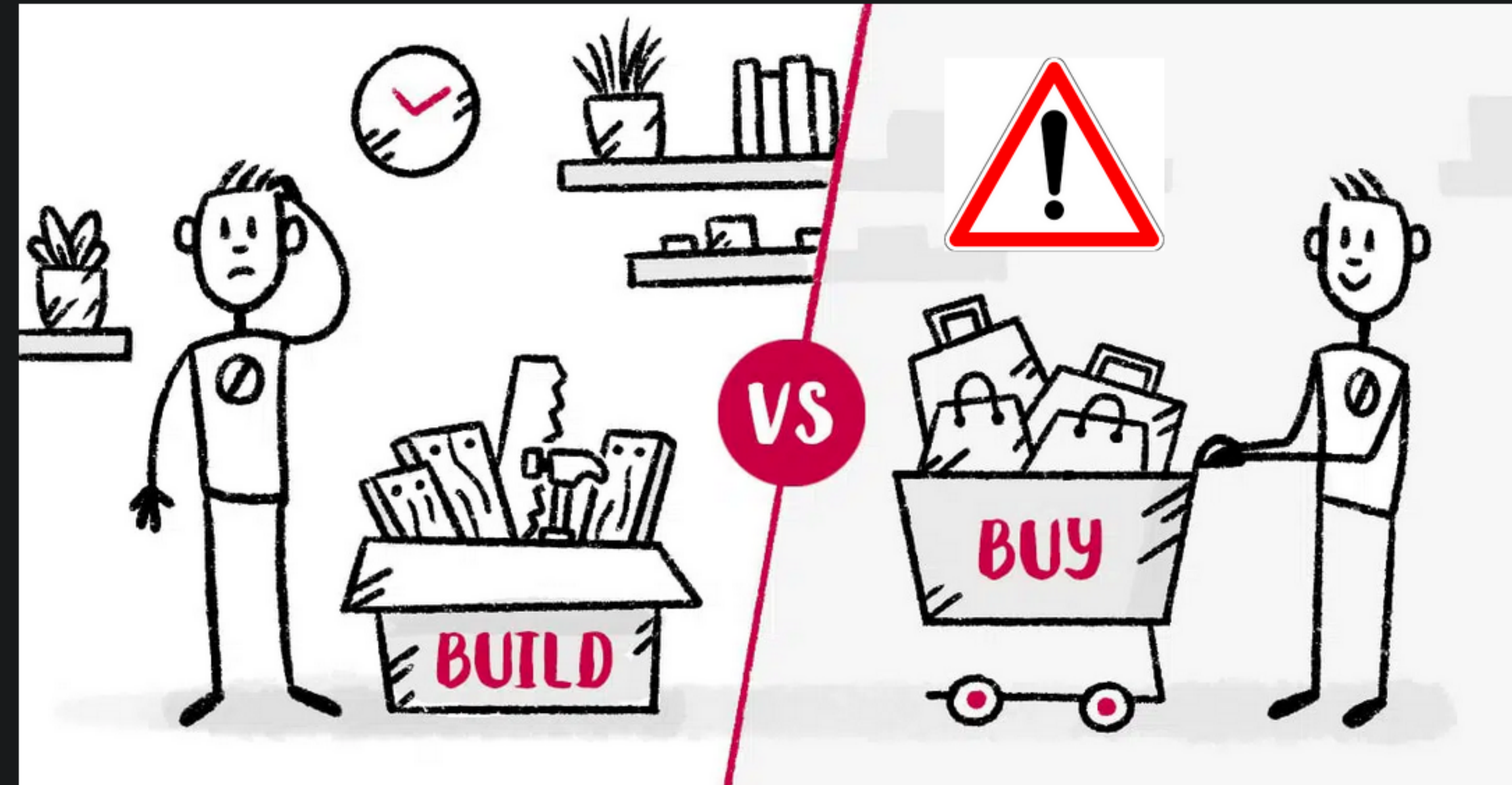
Packaged Software

- Many business needs are **not unique**
 - E.g. payroll, accounts receivable
- Needs for **standardized processes** can be met quite well by packaged software:
 - Ranging from **small single-function tools** (e.g. a client-side file upload manager), to **huge all-encompassing systems** (e.g. ERPs)



!Important!

- Rarely there is a perfect fit.
- You must accept the functionality that is provided by the system.
- This could mean a substantial change in the way the company does business.
- ***Letting technology drive the business can be a dangerous way to go!***



Customization of Packaged Software

- Most packaged software allow for **some customization**:
- If the amount or type of customization is not enough, the project team might create a **workaround**:
 - a **custom-built add-on program** that interfaces with the packaged application **to handle special needs**.
- Workarounds should be the **last resort**:
 - Not supported by the vendor → might become ineffective when the main system is updated
 - If problems arise in the system, the vendors would likely blame the workaround and refuse to provide support.

The search for a software package should be based on the detailed requirements identified during analysis.

I.e. the analysis is still essential: we must know what we need before seeking a product that provides the best fit.

Systems Integration

- Building new systems by combining packaged software, existing legacy system, and new software written to integrate these.
- It is not uncommon for companies to choose the packaged software option and then **outsource** the integration.
 - Many consulting firms specialize in systems integration.
- **Key challenge:** *integrating data*
 - Taking data produced by one package or system and transforming it for use in another package or system

Outsourcing

- Quiet popular
- Hiring an external vendor, developer, or service provider to create or supply the system.
 1. Application Service Provider (ASP):
 - provides applications and/or software related services over corporate networks or the internet.
 - hosts and manages an application on its own servers.
 - bills the customer either on a per-use basis or on a monthly or annual fee basis.
 2. Software as a Service (SaaS):
 - Like ASP, but the vendors develop, manage and provide **their own software**.
 - **Who are some early providers of SaaS?**
 3. Develop a custom system on behalf of the customer

Outsourcing - Custom System

- There are certain risks involved when you decide to leave the creation of a new system in the hands of someone else:
 - You could compromise confidential information
 - You could lose control over future development
 - In-house professionals are not gaining new skills
 - Expertise is transferred to the outside organization
 - Important skills can walk right out the door at the end of the contract
- Two risks are particularly important:
 1. Assess the requirements for the project thoroughly -- **you should never outsource what you do not understand**
 2. **Choose the provider carefully** based on a proven track record with the type of system and technology that your system needs.

Types of Contracts

- **Time and arrangements deal**
 - You agree to pay for whatever time and expenses are needed
 - Very flexible, but could result in a large bill
 - Works best when you and the provider are unclear about what it takes to finish the job.
- **Fixed-price contract**
 - If the provider exceeds the agreed-on price, they have to absorb the costs.
 - You will not pay more than expected
- **Value-added contract**
 - Some percentage of the completed system's benefits are paid to the provider.
 - This is gaining popularity
 - It is quite low-risk if you are willing to share the wealth.

The art of fair contracts

- You need to carefully balance **flexibility** with clearly **defined terms**.
- Keep in mind that **needs often change over time**
- **Short term contracts** leave room for reassessment if needs change or if relationships are not working out the way both parties expected.

- Keep the lines of communication open between you and your outsourcer.
- Define and stabilize requirements before signing a contract.
- View the outsourcing relationship as a partnership.
- Select the vendor, developer, or service provider carefully.
- Assign a person to manage the relationship.
- Do not outsource what you do not understand.
- Emphasize flexible requirements, long-term relationships, and short-term contracts.

FIGURE 6-4 Outsourcing guidelines.

Software Acquisition Options (Pros & Cons)

Pros	Cons
Custom Development	
Get exactly what we want New system built consistently with existing technology and standards Build and retain technical skills and functional knowledge in-house Allows team flexibility and creativity Unique solutions created for strategic advantage	Requires significant time and effort May add to existing backlogs May require skills we do not have Often costs more Often takes more calendar time Risk of project failure
Packages (purchased or obtained from ASP or SaaS)	
No need to “reinvent the wheel” for common business needs Tested, proven product Cost savings Time savings Utilize vendors' expertise Some customization may be possible	Rarely a perfect fit Organizational processes must adapt to software Reliance on vendor for maintenance and future enhancements Will not develop in-house functional and technical skills Unique needs may go unmet May require system integration
Outsourced Development	
Hire expertise we do not have May save time and money Lower risk	No opportunity to build in-house expertise Reliance on vendor Future options limited Security—potential loss of confidential info Performance based on contract terms

FIGURE 6-3 Summary of software acquisition options pros and cons.

How to choose the right strategy?
Should we build, buy or outsource?

Criteria that influence your choice

If the **business need is central** to the company strategy, it is usually better not to outsource, and retain exclusive control over the project if possible.

	When to Use Custom Development	When to Use a Packaged System	When to Use Outsourcing
			
Business need	The business need is unique.	The business need is common.	The business need is not core to the business.
In-house experience	In-house functional and technical experience exists.	In-house functional experience exists.	In-house functional or technical experience does not exist.
Project skills	There is a desire to build in-house skills.	The skills are not strategic.	The decision to outsource is a strategic decision.
Project management	The project has a highly skilled project manager and a proven methodology.	The project has a project manager who can coordinate vendor's efforts.	The project has a highly skilled project manager at the level of the organization that matches the scope of the outsourcing deal.
Time frame	The time frame is flexible.	The time frame is short.	The time frame is short or flexible.

FIGURE 6-5 Selecting a system acquisition strategy.

When skills are not strategic, choosing packaged systems or outsourcing allows internal employees to focus on other business-critical applications and skills.

Seeking additional information

- After understanding how various options might fit with the project needs:
 - What tools and technology would be used if we were to build a custom application in house?
 - What vendors make packaged systems that address the project needs?
 - What service providers would be able to build this system if the application was outsourced?
- To answer these questions:
 - Talk to people in the IS department
 - Get recommendations from business users in other companies with similar needs and ask what they use
 - Look at brochures, product demonstrations and information seminars provided by vendors and consultants.

Seeking Additional Information - RFP

- **Request for Proposal (RFP)**
 - Solicits a formal proposal from a potential vendor, developer, or service provider
 - Describes in detail the system or service that is needed as well as
 - Special technical needs or circumstances
 - Evaluation criteria
 - Procedures to follow
 - Timetable
 - Vendors respond by describing in detail how they could supply those needs and include
 - A schedule and price
 - Vendor proposal is a binding offer
- Once the winning proposal is chosen, a contract is created.

Seeking Additional Information - RFI & RFQ

- **Request for Information (RFI)**
 - Shorter and less detailed than RFP, sent to potential vendors to obtain general information about their products and services
 - Used to see which vendors have the capability and then to follow up with an RFP
- **Request for Quote (RFQ)**
 - When a list of equipments is so complete that the vendor need only provide a price, without any analysis or description of what is needed, the request for quote (RFQ) may be used.

After **evaluating the acquisition strategy** options and **seeking additional information**, the design team will likely have **several viable choices**. Each alternative will have pros and cons and **only one solution can be selected** in the end.

Alternative Matrix

Contains the technical, economical, and organizational **feasibilities** for each candidate, **pros** and **cons**, and **other information**.

Sometimes, **weights** are provided to show when some criteria are more important to the final decision.

Evaluation Criteria	Relative Importance (Weight)	Alternative 1: Custom Application Using VB.NET	Score (1–5)*	Weighted Score	Alternative 2: Custom Application Using Java	Score (1–5)*	Weighted Score	Alternative 3: Packaged Software Product ABC	Score (1–5)*	Weighted Score
Technical Issues:		↑			↑			↑		
Criterion 1	20		5	100		3	60		3	60
Criterion 2	10		3	30		3	30		5	50
Criterion 3	10		2	20		1	10		3	30
Economic Issues:		↓			↓			↓		
Criterion 4	25	Supporting	3	75	Supporting	3	75	Supporting	5	125
Criterion 5	10	Information	3	30	Information	1	10	Information	5	50
Organizational Issues		↓			↓			↓		
Criterion 6	10		5	50		5	50		3	30
Criterion 7	10		3	30		3	30		1	10
Criterion 8	5		3	15		1	5		1	5
TOTAL	100	↓		350	↓		270	↓		360

* This denotes how well the alternative meets the criteria. 1 = poor fit; 5 = perfect fit.

FIGURE 6-6 Sample alternative matrix using weights.

It is entirely possible for an analysts to **skew the analysis** according to whichever alternative they prefer and yet retain the appearance of an objective, rational analysis. To avoid this, each analyst could **develop ratings independently**; then compare and resolve discrepancies in an open team discussion.